Managing Software Quality

CSE 362

Software Development and Professional Practice

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- Quality cannot be added as an afterthought. It has to be built in from the very beginning.

- We will discuss the dimensions of quality of the software product and the software process.

What is Quality?

- ‘Software quality’ is a rather elusive notion.

- Different people will have different perspectives on the quality of a software system.
  - A system tester may view quality as ‘compliance to requirements’, whereas a user may view it as ‘fitness for use’. Both viewpoints are valid, but they may not coincide.

Quality: Measures and Numbers

- Suppose we want to express some quality attribute, say the complexity of a program text, in a single numeric value.

- Larger values are meant to denote more complex programs.

- If a mapping C from programs to numbers can be found, we can compare the values of C(P1) and C(P2) to compare the complexities of programs P1 and P2.

```
procedure bubble
    (var a: array [1..n] of integer; n: integer);
var i, j, temp: integer;
begin
    for i:= 2 to n do
        j:= i;
        while j > 1 and a[j] < a[j-1] do
            temp:= a[j];
            a[j]:= a[j-1];
            a[j-1]:= temp;
            j:= j-1;
        enddo
    enddo
end;
```

```
procedure bubble
    (var a: array [1..n] of integer; n: integer);
var i, j, temp: integer;
begin
    for i:= 2 to n do
        if a[i] ≥ a[i-1] then goto next endif;
        j:= i;
    loop: if j ≤ 1 then goto next endif;
        if a[j] ≥ a[j-1] then goto next endif;
        temp:= a[j];
        a[j]:= a[j-1];
        a[j-1]:= temp;
        j:= j-1;
        goto loop;
    next: skip;
    enddo
end;
```

Taxonomy of Quality Attributes: McCall

Product operation:	
Correctness	Does it do what I want?
Reliability	Does it do it accurately all of the time?
Efficiency	Will it run on my hardware as well as it can?
Integrity	Is it secure?
Usability	Can I run it?
Product revision:	
Maintainability	Can I fix it?
Testability	Can I test it?
Flexibility	Can I change it?
Product transition:	
Portability	Will I be able to use it on another machine?
Reusability	Will I be able to reuse some of the software?
Interoperability	Will I be able to interface it with another system?

Taxonomy of Quality Attributes: ISO 9126

Characteristic	Subcharacteristics
Functionality	Suitability Accuracy Interoperability Security Functionality compliance
Reliability	Maturity Fault tolerance Recoverability Reliability compliance
Usability	Understandability Learnability Operability Attractiveness Usability compliance
Efficiency	Time behavior Resource utilization Efficiency compliance
Maintainability	Analyzability Changeability Stability Testability Maintainability compliance
Portability	Adaptability Installability Co-existence Replaceability Portability compliance

McCall vs. ISO 9126

The ISO 9126 taxonomy places a greater emphasis on user requirements and satisfaction,

while the McCall taxonomy places a greater emphasis on internal software characteristics and development processes.

Bridging the gap between the software developers and the users

- Worry of the developer: tends to focus on the product-base and manufacturing-based definitions of quality.
  - Result: quantifiable terms (e.g. number of defects)

- Worry of the user: tends to focus on rather subjective quality viewpoints, such as 'fitness of use'

Solution: Quality attribute scenarios

Quality attribute: Usability

Source: End user

Stimulus: Learn system features

Artifact: System

Environment: At runtime

Response: Learn tasks supplied by the system for new employees

Response measure: days on the job

Test: 90% successful completion of assigned tasks in employee test for the system, within twice the average time of an experienced user

Worst: 1 to 7 days

Plan: less than 1 day (to passing of test)

Best: less than 2 hours

ISO Standards in Software Engineering

- ISO 9000 is a set of international standards that provide guidelines for establishing, implementing, maintaining, and improving a Quality Management System (QMS) within an organization.

- The ISO 9000 family of standards was first published in 1987 and has since been revised several times, with the latest version being ISO 9001:2015.

- The standards provide a framework for establishing a QMS that covers all aspects of an organization's operations, including management, processes, resources, and customer feedback.

- To become certified to the ISO 9001 standard, an organization must undergo external periodic audits.

How to develop a QMS
Case: Library Management System

- Define the Scope
- Identify Requirements
- Develop Policies and Procedures
- Assign Responsibilities
- Communicate the QMS
- Implement the QMS
- Monitor and Measure
- Continuously Improve

The Capability Maturity Model (CMM)

- The Capability Maturity Model (CMM) provides a framework for evaluating the maturity level of an organization's software development processes

5 levels of the Capability Maturity Model

LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4	LEVEL 5
Initial Software development processes are disorganized.	**Repeatable** Processes are defined and documented.	**Defined** Processes are standardized.	**Managed** Processes are monitored and controlled.	**Optimizing** Processes are continuously improved.

CMM vs. ISO 9000

- The CMM provides a systematic approach for assessing the maturity level of an organization's processes and identifying areas for improvement.

- The ISO 9000 standards provide a framework for implementing a QMS that can help to address these areas.

Exercise: Building a Quality Management Plan

Case Study 1: E-commerce Website Development

- Project Scope: Develop a robust and scalable e-commerce website for a fashion retailer that allows customers to browse, select, and purchase products online.
- Timeline: 9 months
- Budget: $500,000
- Requirements: The website should have the following features: product catalog, shopping cart, payment gateway integration, user authentication and authorization, order management, shipping and tracking, and customer support.

Exercise: Building a Quality Management Plan

Case Study 2: Mobile App Development

- Project Scope: Develop a mobile app for a transportation company that allows users to book rides, track their drivers, and pay for services.
- Timeline: 6 months
- Budget: $200,000
- Requirements: The mobile app should have the following features: user registration and login, ride booking, ride tracking and notification, payment gateway integration, driver tracking and communication, rating and feedback system, and customer support.

Exercise: Building a Quality Management Plan

The plan should include the following:

- Quality objectives: Define the quality objectives of the project. What are the specific quality goals that the project team is trying to achieve?
- Quality standards: Identify the quality standards that will be used to guide the project team in achieving the quality objectives. This may include industry standards such as ISO 9001 or CMMI, or internal standards specific to the organization.
- Quality assurance: Describe the processes that will be used to ensure that the project team is adhering to the quality standards and achieving the quality objectives. This may include reviews, inspections, testing, and other quality control activities.
- Quality control: Define the processes that will be used to measure and monitor the quality of the software product throughout the project lifecycle. This may include metrics, reporting, and defect tracking.
- Quality improvement: Describe the processes that will be used to identify opportunities for quality improvement throughout the project lifecycle. This may include root cause analysis, corrective and preventive actions, and continuous improvement activities.